

Economic complexity Tests and homeworks

Fall 2025

Test – Sept 23

1. What are the assumptions behind the product space?
2. What are the empirical observations made with the use of product space regarding diversification?
3. What are the implications for policy in less developed countries?

Reading and homework – Sept 23

Read: Hausmann and Hidalgo 2009 PNAS.

Homework: Illustrate the dynamics of complexity ranking in trade, technology and research.

1. Go to the country Complexity Rankings on oec.world and download the country data.
2. Investigate ranking dynamics in all dimensions across 2000-2010-2020. Compare correlations across time but within dimensions. Check cross-correlations across dimensions.
3. /Plus: Merge data with Economic Growth forecast tables <https://oec.world/en/resources/methods/eci-growth-predictions>. Calculate the coefficient of ECI indices using the Solow model of economic growth.
4. Prepare visualizations and discuss the results. Collect figures and text to a pdf (max 2 pages) and save it "givenname_familyname_EC2025_HW1.pdf".
5. Send it to the instructor and TA by email. Deadline: October 6, Monday, 8am. Homework sent after the deadline are not considered.
6. Homeworks covering points 1-2-4-5 can get 2.5 points. Point 3 gets an extra 1.5 point.

Test – Oct 7

1. What does it mean that a country ranks high in economic complexity?
2. What does the method of reflections mean?
3. What do the GDP growth regression models using the economic complexity measures tell us?

Reading Oct 7

Audretsch and Feldman 1996 AER

Test Oct 14

Summarize the Audretsch-Feldman article in 3 points.

Your 3 points may include:

- Main messages and findings
- Empirical specifications (variables, regressions)
- Explanations or links to previous classes
- Things you liked, did not like or did not understand

Reading and Homework Oct 14

Read: Neffke et al. 2017 *Journal of Economic Behavior and Organization*

Homework: literature search on the urban phenomena.

1. Identify a topic that is connected to spatial concentration of population, or production, or innovation. This can be important for a city, an industry, a country, or the global humanity (eg. automation, inequality, international competition, climate change, populism etc.).
2. Search for journal articles that address this topic with interesting data. Choose one that you would like to show to the class.
3. Prepare 1 pptx slide that summarizes the paper. Include the research question, the data, method and findings. Try to include a visualization that can tell the story. Save the file: "givenname_familyname_EC2025_HW2.pptx".
4. Send it to the instructor and TA by email. Deadline: October 20, Monday, 8am. Homework sent after the deadline are not considered.
5. Homeworks covering 1-2-3-4 can get 2.5 points. 3-5 homeworks can be presented during the Oct 21 class to collect 1 „Activity point” each .